1 What drives it, besides DHT?

- Bald vs haired scalp, 2 studies (GEO)
- DHT on follicle cells (GEO)
- Open Targets genetics

Scarring (TGF-beta) program, survives the follicle-loss adjustment

2 Does a drug reverse it?

- Fibroblasts with TGF-beta and pirfenidone at 0.11 and 1 mM (GEO)
- CXCL12 check (2025 mouse study)

Pirfenidone reverses it at 1 mM and cuts CXCL12 3.8x

3 Right cells, right people?

- Scalp single-cell atlas
- Beard vs scalp, one man
- FDA reports, 20.7M (FAERS)

Sheath, papilla and fibroblasts; 70% fewer hair-loss reports than nintedanib

4 Does it hold up?

- Baldness GWAS + eQTL
- LLM evidence agent, blind-audited
- Community scan; dose check

Mechanism Strong in the literature; genetics weak; dose uncertain; untested

Negative results reported alongside: drug-signature screening failed calibration; 92% of the non-hormonal genetic targets have no drug; blood-pressure drugs (ARBs) dropped; the genetic natural experiment did not confirm the pathway.